

Original Research

Heavy Metal Pollution in Soils of the Oilfield-Wetland Zone in the Yellow River Delta: Distribution, Source Analysis, and Ecological Risk Assessment

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Abstract

Oilfield-wetland zones are ecologically fragile landscapes where petroleum extraction, traffic, and agricultural activities may jointly affect soil metal accumulation. In this study, surface soils from oilfield-wetland patches in the Yellow River Delta were analyzed for Cr, Co, Ni, Cu, Zn, Cd, and Pb, together with selected soil properties. The concentrations of Cr, Co, Ni, Cu, Zn, Cd, and Pb ranged from 47.11 to 106.72, 11.01 to 23.53, 27.61 to 64.28, 6.41 to 60.45, 10.41 to 190.95, ND to 0.37, and 16.53 to 104.58 mg/kg, respectively, and were below the Chinese agricultural soil risk-screening values. Ni exceeded the local background value in all samples, while Pb, Co, Cr, and Cu showed frequent background exceedance, indicating limited enrichment rather than regulatory risk. Spatial differences were element-specific, with Cu differing significantly between Gudao and Xianhe towns and Zn tending to be higher away from oil wells and in vegetated zones. Correlation and PCA results suggested possible influences from both natural processes and anthropogenic activities. Pollution and ecological risk indices indicated low overall risk, but Pb enrichment, consistent Ni background exceedance, and localized Cd-related ecological risk require continued monitoring.

Keywords: heavy metals, oilfield, Yellow River Delta, spatial distribution, potential sources

Introduction

With the continuous expansion of urbanization and industrialization, environmental pollution has received

increasing attention because of its adverse effects on human health and quality of life [1]. Heavy metals, unlike most organic compounds, are inherently resistant to degradation in different environmental matrices and can accumulate continuously within food chains and webs, ultimately posing severe health threats to plants, animals, and humans [2]. Heavy metals can enter plants through root systems, be transported via the xylem

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in ionic or complexed forms, and ultimately accumulate in shoot tissues, causing growth inhibition, necrosis and other toxic effects [3, 4]. They can also be transferred to the next trophic level through feeding behavior, and stable isotope-based studies have been used to examine heavy metal accumulation in marine organisms [5, 6]. For instance, a study analyzing fecal samples from insectivorous bats inhabiting different habitats found that bats from mining areas accumulated higher heavy metal concentrations than those from farmland habitats [7]. A number of studies have shown that Cd accumulation in crops varies substantially among plant species, cultivars, and soil conditions [8]. High exposure to heavy metals may impair physiological and immune functions in aquatic organisms [9], and may also cause malignant diseases, renal dysfunction, and central nervous system disorders in humans, some of which are difficult to fully reverse with current medical interventions [10, 11].

Human activities are widely recognized as the primary contributors to heavy metal pollution. These activities include mining operations [12], improper use of pesticides and chemical fertilizers [13], discharge of industrial and agricultural wastewater [14, 15], and transportation exhaust emissions [16]. Oilfield exploitation may be an important source of local metal pollution since crude oil contains multiple heavy metals and extraction may lead to elevated heavy metal concentrations in soils nearby [17]. For example, a previous study in the Yellow River Basin showed that soil heavy metal concentrations were slightly higher in areas proximate to oil and gas development zones compared to more distant regions [18]. Subsequent oilfield-related activities, such as crude oil transportation, can further propagate heavy metal pollution, posing potential threats to local ecosystems [16]. Therefore, it is imperative to intensify investigations into heavy metal pollution around oilfields to provide more precise guidance for subsequent planning and remediation efforts targeting environmental heavy metal contamination.

The Yellow River Delta (YRD) region is one of the largest river deltas in China and also ranks among the world's most dynamic areas for terrestrial-marine ecosystem interactions as well as serves multiple economic functions as a vital energy base and chemical industry hub for China [19]. Plentiful wetland resources are another hallmark of the YRD, and the sediment transport and land-building processes of the Yellow River drive the annual formation of extensive newly formed wetlands in this area [20]. The soil types are predominantly saline-alkali soils and coastal alluvial soils, with vegetation dominated by alkali-tolerant plants, resulting in a relatively fragile ecological environment [21]. Changes in water and sediment quality can also affect riparian vegetation, showing the close link between environmental conditions and organisms in wetland ecosystems [22]. Shengli Oilfield, one of the earliest-developed large-scale oilfields in China, is located within this region, with its main production area situated in Dongying City in the lower

reaches of the Yellow River. Elevated levels of As, Cd, and Pb were observed in environmental media in the wetlands in the YRD due to long-term impacts from oil and gas industrial activities [23]. Previous studies have also reported persistent organic pollutants in aquatic products from the Yellow River Delta [24], suggesting the contaminant-accumulation potential of this deltaic ecosystem. So far, several studies have been conducted focusing on heavy metal pollution in natural or human-affected wetland ecosystems in the YRD [23, 25-28]. However, limited attention has been paid to oilfield-wetland coupled zones, where oil well pads are embedded within fragile wetland landscapes and may interact with wetland biogeochemical processes. How oil well proximity and vegetation heterogeneity modulate heavy metal accumulation in these coupled systems remains poorly understood. Integrating distance-to-well gradients and vegetation contrasts could provide additional evidence on how oilfield activities may interact with wetland conditions to affect heavy metal accumulation, thereby refining the current understanding of contamination patterns in the YRD. Given these circumstances, it is necessary to investigate heavy metal contamination in these particularly fragile zones, analyzing the distribution patterns, potential sources and potential risks of heavy metals to support ecological conservation efforts and the formulation of relevant land management policies in the YRD. Thus, in this study, surface soils in oilfield-wetland zones in the YRD were collected to 1) measure the concentrations of typical heavy metals; 2) reveal the spatial distribution of heavy metals; 3) explore potential sources and evaluate corresponding ecological risks of heavy metals.

Materials and Methods

Sampling

A total of 14 wetland patches containing active oil wells in Gudao Town and Xianhe Town in Dongying City, which are core oil-producing areas of the Shengli Oilfield and are characterized by the coexistence of intensive petroleum infrastructure and wetland ecosystems, were randomly selected as sampling sites (Fig. 1). Surface soil is closely associated with the plant root zone and easily affected by the direct impacts of surface deposition and/or some oilfield operations. Surface soil is commonly used to evaluate the ecological risks of pollutants. And in this study, surface soil samples (0-20 cm) were collected ($n = 39$; each was a composite of 5 sub-samples from the sampling site), with stratification by administrative location (23 from Gudao Town, 16 from Xianhe Town), distance (12 samples near wells (<10 m), 27 samples far from wells (>10 m)), and vegetation cover (22 vegetated, 17 non-vegetated). To some extent, the sampling ensured representative coverage of the major environmental gradients in the oilfield-wetland system. All soil samples were

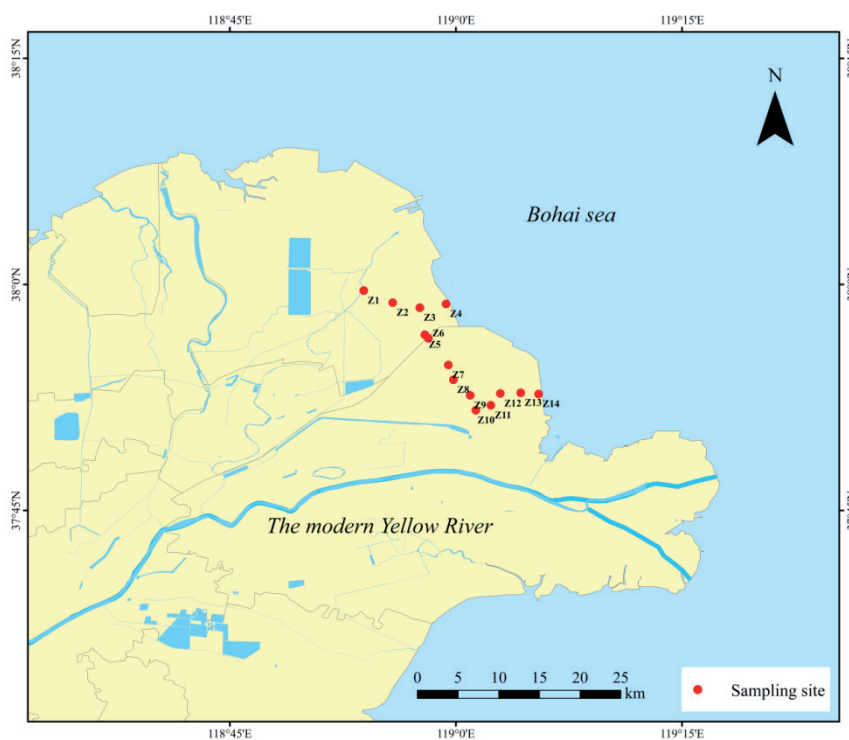


Fig. 1. Simplified map of the study area and sampling locations.

placed in polyethylene bags and stored at -20°C until subsequent analysis. The collected soils were classified as Kenli shoal sandy soil, a typical coastal sandy soil type characterized by a sandy texture with high sand content, low clay fraction, and loose soil structure in the Yellow River Delta. Information on the physicochemical properties of the soils is shown in Table 1.

Sample Analysis

Soil samples were pre-treated following the method of Zhang et al. [29] with minor modifications. Visible debris like stones, shell fragments and plant/animal residues were manually removed. Then the samples were freeze-dried to constant weight in a vacuum freeze-dryer (Scientz-10ND, Xinzhi, Ningbo, China) for 72 h. The dried samples were gently ground and passed through a 0.1 mm nylon sieve. After sieving, 0.15 g of each sample was microwave-digested (Ethos Up, Milestone, Sorisole, Italy) using an $\text{HNO}_3\text{-HCl-H}_2\text{O}_2$ system at 185°C for 45 min. The acids in the digested mixtures were evaporated through heating. Then, the digestion solution was heated to remove excess acid mist. The residues were diluted to a specified volume by adding 5% HNO_3 . Concentrations of metal elements including Cr, Co, Ni, Cu, Zn, Cd, and Pb were measured by ICP-MS (iCAP-RQ, Thermo Scientific, America). The concentrations of targeted elements were uniformly expressed as mg/kg (dry weight). For each batch, a reagent blank, a sample blank, a reference material (ERM-S-510204, TMRM, Changzhou, China) and duplicate samples were incorporated. The calibration

curves for the targeted elements showed excellent linearity ($R^2 > 0.9996$) over the concentration range of 0.01–10 mg/L. The recoveries of the reference material were above 85%, 82%, 91%, 95%, 92%, 79% and 87% for Cr, Co, Ni, Cu, Zn, Cd, and Pb, respectively. The relative standard deviations of elements for parallel samples were less than 6%. Values below the detection limit were treated as zero for statistical analysis.

Basic physico-chemical properties including pH, particle size and total organic carbon (TOC) were measured based on standard methods. Soil pH was measured in a 1:5 soil-water mixture using a glass electrode method. TOC was measured according to the $\text{K}_2\text{Cr}_2\text{O}_7$ volumetric method. Particle size distribution was determined using a hydrometer method. Detailed procedures can be found in our previous study [29].

Pollution Degree and Ecological Risk Assessment

To evaluate the pollution status and potential ecological risks of heavy metals in wetland zones affected by oil wells, several typical methods were used. The single-factor pollution index was used to identify element-specific exceedance, while the Nemerow composite index was applied to characterize the integrated contamination status of heavy metals at each site. The geoaccumulation index was employed to evaluate the magnitude of anthropogenic accumulation relative to the natural geochemical background. The potential ecological risk index (R_I) was adopted to assess the ecotoxicological hazard of the detected elements.

Table 1. Concentrations of heavy metals (mg/kg) and soil properties in oilfield-wetland soils.

	Cr	Co	Ni	Cu	Zn	Cd	Pb	pH	Clay (%)	TOC (%)
Minimum	47.1	11.0	27.6	6.4	10.4	ND	16.5	7.0	3.5	0.2
Maximum	106.7	23.5	64.3	60.5	191.0	0.4	104.6	8.6	23.8	21.4
Mean	69.3	15.6	39.9	23.4	56.8	0.1	37.4	7.6	11.1	5.7
Median	67.0	16.0	39.5	22.1	38.8	0.04	31.44	7.66	9.76	4.11
CV	0.18	0.17	0.20	0.44	0.80	1.16	0.55	0.04	0.38	0.68
Background Value	65.5	11.4	27.5	21	62.2	0.13	19.4			
GB 15618-2018	250	-	190	100	300	0.6	170			

Note: ND indicates not detected. ND values were treated as zero for statistical analysis and index calculation.

The single pollutant index, described as $P_i = C_i/S_i$ (P_i , C_i and S_i represent the pollution index of a single element i , the measured concentration of element i and the associated evaluation standard for metal i), was calculated to evaluate the pollution level of a single metal element in the soils. S_i values for Cr, Ni, Cu, Zn, Cd and Pb were 250, 190, 100, 300, 0.6, and 170 mg/kg based on the risk screening values in the risk control standard for soil contamination of agricultural land in China [30, 31]. Co was not included in the P_i and Nemerow index calculations because no corresponding screening value was available in GB 15618-2018. The pollutant levels were listed as not contaminated ($P_i \leq 1$), mildly contaminated ($1 < P_i \leq 2$), moderately contaminated ($2 < P_i \leq 3$) and heavily contaminated ($P_i > 3$) [31].

The Nemerow comprehensive pollution index was calculated to reflect the overall pollution status of pollutants, specifically highlighting the impact of the most severely polluting factors [32]. It was described as

$$P_{int} = \sqrt{\frac{(P_{i \max}^2 + P_{i \text{ave}}^2)}{2}}$$

where P_{int} denotes the Nemerow composite pollution index; $P_{i \max}$ is the maximum value of P_i of heavy metals at a given sampling point; $P_{i \text{ave}}$ is the average value of P_i of heavy metals at a given sampling point. The P_{int} ranges of ≤ 0.7 , 0.7-1, 1-2, 2-3 and > 3 corresponded to a safe/clean contamination level, the warning threshold, mild contamination, moderate contamination and severe contamination, respectively [33].

The geoaccumulation index (I_{geo}), which compares measured metal concentrations with geochemical background values, was calculated to further assess the pollution degree of heavy metals in soils [34]. The calculation formula was as follows:

$$I_{geo} = \log_2 \frac{C_n}{1.5B_n}$$

Here, the specific background values used for Cr, Co, Ni, Cu, Zn, Cd, and Pb were 65.5, 11.4, 27.5, 21.0,

62.2, 0.126, and 19.4 mg/kg, respectively. And seven categories were defined as ≤ 0 (uncontaminated), 0-1 (mildly contaminated), 1-2 (moderately contaminated), 2-3 (moderately to highly contaminated), 3-4 (highly contaminated), 4-5 (highly to extremely highly contaminated), and > 5 (extremely highly contaminated) [35].

The ecological risk index method was used to assess the potential ecological hazard posed by a specific element and multiple metals in soils within a given area. Related formulas were as follows:

$$E_r^i = T_r^i \cdot c_s^i / c_n^i$$

$$RI = \sum E_r^i$$

where E_r^i represents the potential ecological risk coefficient of metal i in soil. T_r^i is the toxicity coefficient of element i , whose values for Cr, Co, Ni, Cu, Zn, Cd and Pb were 2, 5, 5, 5, 1, 30 and 5, respectively [36]. Here, c_s^i and c_n^i denote the measured concentration and the reference value of metal (i), respectively. The c_n^i values for Cr, Co, Ni, Cu, Zn, Cd and Pb were adopted from the local geochemical background values listed in Table 1 [37]. RI denotes the integrated potential ecological risk index of multiple metals in soils. Based on the values of E_r^i and RI , risk levels were classified as slight ecological risk ($E_r^i < 40$ or $RI < 150$), moderate ecological risk ($40 \leq E_r^i < 80$ or $150 \leq RI < 300$), considerable ecological risk ($80 \leq E_r^i < 160$ or $300 \leq RI < 600$), very high ecological risk ($160 \leq E_r^i < 320$ or $RI \geq 600$) and extremely high ecological risk ($E_r^i \geq 320$) [36].

Statistical Analysis

Statistical analysis was performed using SPSS 19.0 (IBM, USA), with the significance level set at $\alpha = 0.05$. Data normality was assessed using the Shapiro-Wilk test. For comparisons between two groups, including the differences in heavy metal concentrations between the two towns, an independent-samples t-test was employed when the data met parametric assumptions; otherwise, the Mann-Whitney U test was used. For comparisons

among more than two groups, one-way ANOVA was applied when parametric assumptions were met, whereas the Kruskal–Wallis H test was used when these assumptions were not met. The corresponding test statistics and p-values were reported where significant differences were observed. Spearman's rank correlation analysis was used to examine relationships among heavy metals and soil physicochemical properties. Principal component analysis (PCA) with varimax rotation was used to examine associations among the heavy metals and provide clues to their possible sources.

Results and Discussion

Levels of Heavy Metals in Soils of the Oilfield-Wetland Zone

Concentrations of heavy metals and physico-chemical properties of soils of the oilfield-wetland zones are shown in Table 1. The concentrations of Cr, Co, Ni, Cu, Zn, Cd, and Pb ranged from 47.1 to 106.7, 11.0–23.5, 27.6–64.3, 6.4–60.5, 10.4–191.0, ND–0.4, and 16.5–104.6 mg/kg, respectively. Based on the average levels of those elements, an overall enrichment order of $\text{Cr} > \text{Zn} > \text{Ni} > \text{Pb} > \text{Cu} > \text{Co} > \text{Cd}$ could be observed. Compared with the background values of trace elements in soils of Dongying City (Table 1; [37]), Ni concentrations in all the samples were above the corresponding background value. Besides, concentrations of Co, Pb, Cr and Cu were observed to

exceed the corresponding background values in over 50% of the samples (94.9%, 92.3%, 59% and 51.3% for Co, Pb, Cr, and Cu, respectively). Based on the mean or median values, the concentrations of Zn and Cd were in line with their corresponding local background levels, which was consistent with the findings reported by Yang et al. [25] for soils in the YRD wetlands. This may indicate limited enrichment relative to local background values, although both natural parent materials and anthropogenic inputs may contribute to the observed patterns. On the other hand, from an agricultural management perspective, all measured concentrations were below the risk screening values for agricultural soils in China [30], indicating that no regulatory exceedance or contamination in the regulatory context exists, and suggesting that heavy metals are unlikely to limit agricultural soil use.

Previous studies in the Yellow River Delta also reported considerable spatial variation in heavy metal concentrations among different soil and land-use settings [38, 39], as summarized in Table 2. Comparison with data on heavy metal pollution from other delta regions revealed that the Pb content in this study was relatively higher than that reported in the Qionghai Lake wetland in Southwest China [40] and the coastal delta wetland of the Lower Meghna Estuary in Bangladesh [41]. During the crude-oil exploration processes, the use of drilling fluids and additives (e.g. barite) in petroleum operations might lead to the release of heavy metals into the environment during drilling [42]. Equipment wear, lubricant leakage, and emissions from industrial

Table 2. Heavy metal concentrations (mg/kg) in soils from different wetlands or oil field areas.

Region	Cr	Co	Ni	Cu	Zn	Cd	Pb	References
Yellow River Delta, China	69.3	15.6	39.9	23.4	56.8	0.1	37.4	This study
Yellow River Delta, China	70.3	11.2	27.5	20.7	65.2	0.1	19.0	[25]
Yellow River Delta, China	62.9	-	-	19.6	57.5	0.1	19.4	[23]
Yellow River Delta, China	56.6	-	26.3	18.4	78.2	0.2	20.8	[27]
Yellow River Delta, China	59.7	-	40.0	26.4	111.1	0.6	34.4	[28]
Yellow River Delta, China	122.0	-	30.1	23.0	43.7	0.36	21.8	[38]
Yellow River Delta, China	58.55	-	26.11	12.89	55.97	0.13	17.04	[39]
Qionghai Lake Wetland, China	88.9	-	27.2	28.6	90.3	0.8	18.3	[40]
Meghna River Estuarine Delta, Bangladesh	36.6	-	38.8	35.7	62.3	-	7.4	[41]
Abteymour Oilfield, Iran	126.3	-	77.5	26.1	51.9	-	13.2	[43]
Ahwaz Oil Field, Iran	107.7	-	46.2	56.2	-	0.4	490.8	[45]
Auca Oilfield, Ecuador	-	-	-	570.0	21.0	0.1	2.9	[46]
Matruh Governorate, Egypt	1.8–6.1	0.5–6.5	-	5.6–24.6	2.6–23.3	0.5–6.5	3.8–52.1	[47]
Bodo, Nigeria	0.0–0.9	-	0.61–4.8	0.3–13.41	-	<0.001	<0.001	[48]

Note: Values represent means or ranges as reported in the original studies; “-” indicates that the corresponding element was not reported or not available in the original study.

processes associated with petroleum extraction might also cause metal pollution in the oilfield areas [43]. By contrast, the concentrations of Cu, Zn, and Cd in the study area were much lower than those found in sediments impacted by shipbreaking activities such as historical use of nickel-copper alloys in key ship components (e.g. crankshafts, turbine blades) in regions with well-developed shipping industries [44].

When compared with heavy metal profiles in major oil-producing regions, the heavy metal profile in this study also showed distinct differences. For instance, relatively higher levels of Ni and Cr were reported in data from Iran's Abteymour Oilfield [43] and Ahwaz Oilfield [45]. Much higher concentrations of Cu were reported in soils from Oca Oilfield in Ecuador [46]. Additionally, the Cd concentration in this study was slightly lower than that in oil extraction areas of Matrouh Governorate in Egypt [47] but relatively higher than levels reported in relevant studies from Nigeria [48].

Overall, these global comparisons indicate that heavy metal contamination in oilfield-associated soils varies considerably among regions [43, 45-48]. Such differences may be related to crude-oil composition, production intensity, drilling additives, spill history, local parent materials, soil properties, and surrounding land-use patterns [42, 46, 49]. Compared with heavily contaminated oil-producing areas such as parts of Iran, Ecuador, Egypt, and Nigeria [43, 45-48], the oilfield-wetland soils in the present study showed relatively low concentrations of most metals under the Chinese agricultural soil risk-control framework. This suggests that oilfield activities in the study area may have contributed limited or localized metal inputs rather than causing severe regional contamination. Meanwhile, the consistent exceedance of Ni above the local background value and the localized enrichment of Pb and Cd indicate that long-term monitoring remains necessary in oilfield-wetland coupled zones.

Distribution Pattern of Heavy Metals

Firstly, Fig. 2a) shows the variations in metal concentrations between sites near the oil wells and those at a distance. The results indicated that most elements showed no obvious distributional differences between the two groups, except for Zn. Zn levels were generally higher at sites farther from the oil wells. This pattern suggested that oil exploration was not the major factor governing heavy metal spatial distribution at the wetland patch scale. The elevated Zn levels in areas distant from the oil wells may be linked to more frequent human agricultural activities. This explanation was consistent with previous studies in the Yellow River Delta region, which reported relatively high levels of Zn [25-27]. For instance, Fu et al. [26, 27] observed that Zn content increased with distance from urban centers, potentially due to the historical application of pesticides and fertilizers in cotton cultivation. Notably, sites farther from the oil wells often supported wetland vegetation

such as reeds and seepweeds. Similarly, as shown in Fig. 2b), no statistically significant differences in heavy metal concentrations were observed between vegetated and non-vegetated samples. However, Zn tended to be marginally higher in vegetated samples than in non-vegetated samples, although the difference did not reach statistical significance according to the Mann-Whitney U test ($U = 120.000$, $Z = -1.898$, $p = 0.058$). This tendency may be associated with additional human activities in vegetated areas, such as reed harvesting and

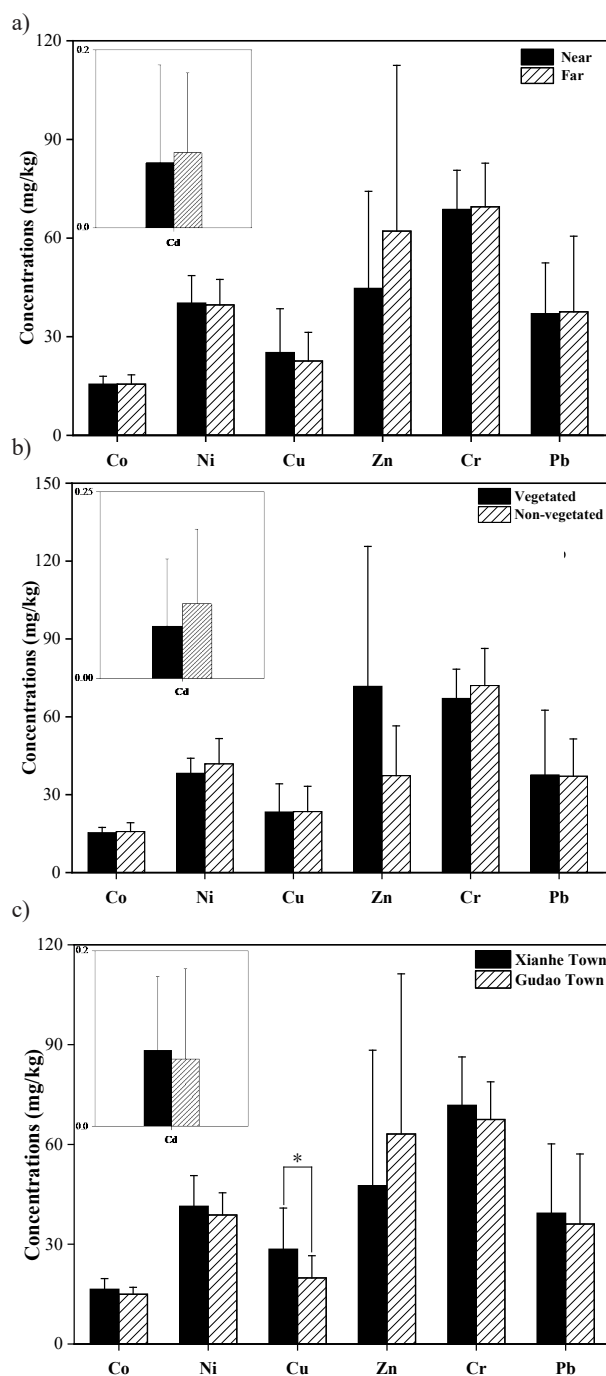


Fig. 2. Spatial distribution patterns of heavy metals in soils of the oilfield-wetland zones, Yellow River Delta: a) distribution across distance groups; b) distribution across vegetation groups; c) spatial variation between Gudao Town and Xianhe Town.

related soil disturbance, although the difference was not statistically significant.

The spatial variation of heavy metals in soils across the two regions, Gudao Town and Xianhe Town, is shown in Fig. 2c). The results indicated that Cu was the only element exhibiting a significant difference in content between the two towns. Because the homogeneity of variance assumption was not satisfied according to Levene's test ($p = 0.043$), an unequal-variance independent-samples t-test was used. The Cu concentration differed significantly between the two towns ($p = 0.019$). Sampling sites in Xianhe Town were relatively dispersed and had additional industrial activities (e.g. aquaculture, chemical industry), whereas those in Gudao Town were more concentrated and had no other obvious activities besides oil exploration. The observed variation in Cu may be associated with differences in parent materials, land-use patterns, and local industrial activities between the two towns [49]. Besides, some factors such as land management practices and hydrological processes might also contribute to the observed heavy metal spatial patterns [50-52], and these issues warrant attention in future monitoring work.

Correlation Among Heavy Metals and Soil Properties

The results of the correlation analysis among heavy metals and soil physicochemical properties were presented in Table 3. A significant inter-correlation was observed among Cr, Co, Ni, Cu, Cd, and Pb, with the strongest correlations being Co-Ni (0.871), Cr-Co (0.782), Co-Cu (0.798), and Cu-Pb (0.725). This strong positive correlation may indicate similar geochemical behavior and potential common sources for Cr, Co, Ni, Cu, and Cd. Their mean and median concentrations were close to the local geochemical background values, suggesting

a predominantly natural pedogenic origin with potential subtle anthropogenic inputs from petroleum exploration activities (e.g. drilling fluid leakage, equipment wear and crude oil transportation). Such slight anthropogenic inputs may occur in long-exploited oilfield-wetland zones and cannot be completely excluded despite the similarity between measured concentrations and background values, especially for Cr and Cu, which may also be influenced by petroleum-related activities. In contrast, Zn showed only a marginal correlation with Cr, suggesting a comparatively different distribution pattern. This distinct behavior was likely related to anthropogenic sources such as fuel additives and legacy pesticides.

The significant positive correlations among Cr, Co, Ni, Cu, Cd, and Pb ($r = 0.457 - 0.871$, Table 3) indicated possible similarities in geochemical behavior and similar environmental behaviors. From a geochemical perspective, the observed correlations may partly reflect shared interactions of these metals with soil minerals and organic matter [53-56]. The particularly strong correlation between Ni and Co ($r = 0.871$) suggested that these two elements may have been influenced by similar geochemical controls. For Cr and Cu, their strong association ($r = 0.713$) may be related to their similar affinity for soil organic matter – both can form stable inner-sphere complexes with functional groups (e.g., $-COOH$, $-OH$) in humus, resulting in co-accumulation in the soil [55,56]. In addition, Pb and Cu ($r = 0.725$) tend to co-precipitate with carbonate minerals (e.g. calcite) under the weakly alkaline conditions ($pH > 7.0$) of the study area, which may enhance their geochemical correlation in the Kenli shoal sandy soil [57].

In contrast, TOC showed no significant correlations with heavy metal concentrations. Although previous studies have shown that nano-humic materials can adsorb heavy metals in soil, this effect was not evident at this study scale [58]. However, the high coefficient of

Table 3. Spearman correlation coefficients among heavy metals and soil physicochemical properties in surface soil samples.

	Cr	Co	Ni	Cu	Zn	Cd	Pb	Clay	pH	TOC
Cr	1									
Co	0.782**	1								
Ni	0.700**	0.871**	1							
Cu	0.713**	0.798**	0.721**	1						
Zn	0.326*	0.184	0.031	0.153	1					
Cd	0.683**	0.603**	0.662**	0.483**	0.187	1				
Pb	0.690**	0.621**	0.540**	0.725**	0.197	0.457**	1			
Clay	0.353*	0.334*	0.329*	0.299	0.041	0.18	0.417**	1		
pH	0.227	-0.068	-0.003	-0.038	0.191	0.369*	0.056	0.091	1	
TOC	0.109	0.068	0.049	0.145	-0.133	0.118	0.055	0.29	-0.083	1

Note: Values represent Spearman correlation coefficients. * $p < 0.05$; ** $p < 0.01$.

variation for TOC (0.68) did suggest substantial spatial heterogeneity in soil organic matter. Such heterogeneity may partly reflect local differences in vegetation density and nutrient conditions [59].

Cd showed a significant positive correlation with soil pH (Table 3), which was closely related to the chemical speciation transformation of Cd in the weakly alkaline Kenli shoal sandy soil. Within the pH range of 7.0–8.63 in the study area, Cd^{2+} in the sandy soil solution could readily react with carbonate ions (CO_3^{2-}) and hydroxide ions (OH^-) generated by soil hydrolysis to form insoluble CdCO_3 and $\text{Cd}(\text{OH})_2$ precipitates [60]. Meanwhile, the increase in pH enhanced the negative charge density on the surface of limited soil colloids (clay minerals, organic matter) in the shoal sandy soil, which further strengthened the electrostatic adsorption of Cd^{2+} [54]. The dual effect of precipitation and adsorption immobilized Cd in the alkaline shoal sandy soil, thus forming a significant positive correlation between Cd concentration and soil pH [61].

Principal component analysis (PCA) was employed to explore the potential sources of heavy metals (Table 4; Fig. 3). Two principal components (eigenvalues >1) were extracted that collectively explained 70.53% of the total variance, capturing the primary data structure for source identification.

PC1 accounted for 53.49% of the total variance and showed high loadings for Cr, Co, Ni, Cu, and Cd, indicating similar variation patterns among these metals. The mean concentrations of Cr, Co and Cu were close to the local background values, suggesting that natural pedogenic processes may have played an important role in their distribution. The slight exceedances in some samples (e.g., Cr in 59% of samples, Cu in 51.3% of samples) implied minor anthropogenic inputs from petroleum exploration activities, such as the use of barite in drilling fluids and equipment abrasion during oil extraction. The consistent enrichment of Ni in all samples, together with its mean concentration exceeding the local background value, may indicate an additional contribution from petroleum-related activities, although the influence of parent materials cannot be excluded [26, 27]. Although the mean concentration of Cd was slightly below the background value, its high coefficient

of variation and elevated concentrations at individual sampling sites may indicate a localized influence, potentially related to petroleum-related activities. The loading of Cd in PC1 indicated its covariance with other heavy metals under mixed natural and anthropogenic influences.

PC2 (17.04% of variance) was mainly characterized by Zn and, to a lesser extent, Pb, and may represent a mixed traffic-agricultural source (Table 4). In the rotated component matrix, Zn exhibited a high loading on PC2 (0.906), suggesting a distinct variation pattern that may be associated with anthropogenic activities. In the Xianhe and Gudao towns of Dongying, the long-term application of phosphate fertilizers and manures to surrounding cotton and paddy fields may have contributed to Zn and Pb inputs, whereas Pb-containing pesticide residues may have enhanced the Zn-Pb covariation [62, 63]. Although small amounts of Zn could be released from drilling-fluid additives and equipment abrasion during petroleum exploitation [42, 46], the trunk roads S310 and S315 carried heavy traffic of oilfield service vehicles and heavy-duty trucks; therefore, tire and brake wear may represent potential sources of Zn and Pb [64]. Pb showed similar loadings on PC1 (0.522) and PC2 (0.533), suggesting a composite origin: the study area contained numerous oilfield well pads, while several sampling sites were adjacent to major roads and farmland. The comparable loadings of Pb on PC1 and PC2 suggest that Pb may have been influenced by multiple sources rather than a single dominant source.

Assessment of Heavy Metal Pollution

The results of the single-factor pollution index and the Nemerow composite pollution index for the soil samples are presented in Fig. 4a). The P_i values for Cr, Ni, Cu, Zn, Cd, and Pb ranged from 0.19 to 0.43, 0.15 to 0.34, 0.06 to 0.61, 0.04 to 0.64, 0 to 0.61, and 0.10 to 0.62, respectively. All P_i values were below 1, indicating an uncontaminated status for individual metals according to the single-factor pollution index. Furthermore, the composite pollution index values in the area ranged from 0.16 to 0.51, below the 0.7 limit.

Table 4. Principal component analysis results and rotated component loadings of heavy metals in surface soils

Principal component	Eigenvalues	Variance (%)	Cumulative variance (%)	Element	PC1	PC2
1	3.744	53.485	53.485	Cr	0.852	0.342
2	1.193	17.041	70.526	Co	0.896	0.12
3	0.792	11.317	81.843	Ni	0.929	-0.148
4	0.650	9.286	91.128	Cu	0.697	0.161
5	0.351	5.021	96.149	Zn	-0.103	0.906
6	0.176	2.510	98.660	Cd	0.696	-0.089
7	0.094	1.340	100.000	Pb	0.522	0.533

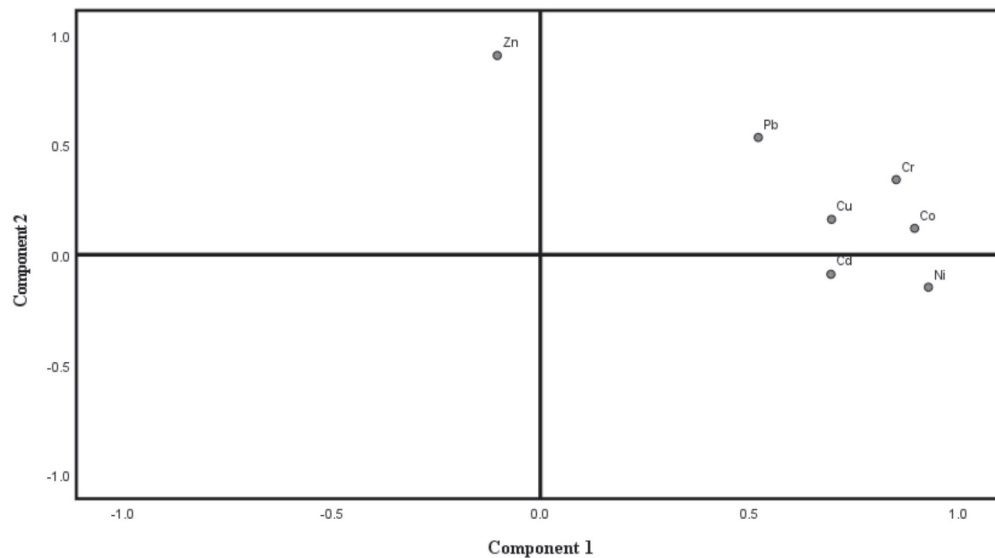


Fig. 3. Factor analysis loading plot for heavy metals in surface soils of the oilfield-wetland zone, Yellow River Delta.

These results suggest a low overall pollution level under the adopted assessment framework.

As shown in Fig. 4b), the I_{geo} values for Cr were below 0 in 97.4% of the samples, whereas 2.6% were between 0 and 1, indicating predominantly uncontaminated conditions with localized mild Cr enrichment. Zn had the lowest median I_{geo} value, with 5.1% of the values between

1 and 2 and 17.9% between 0 and 1, suggesting localized mild to moderate Zn enrichment. Regarding Pb, 12.8% of its index values ranged from 1 to 2, and a significant 48.72% were between 0 and 1, indicating a certain degree of Pb pollution in the study area. Furthermore, a recent study in the western Yellow River Delta reported lower Pb concentrations and generally low ecological risks,

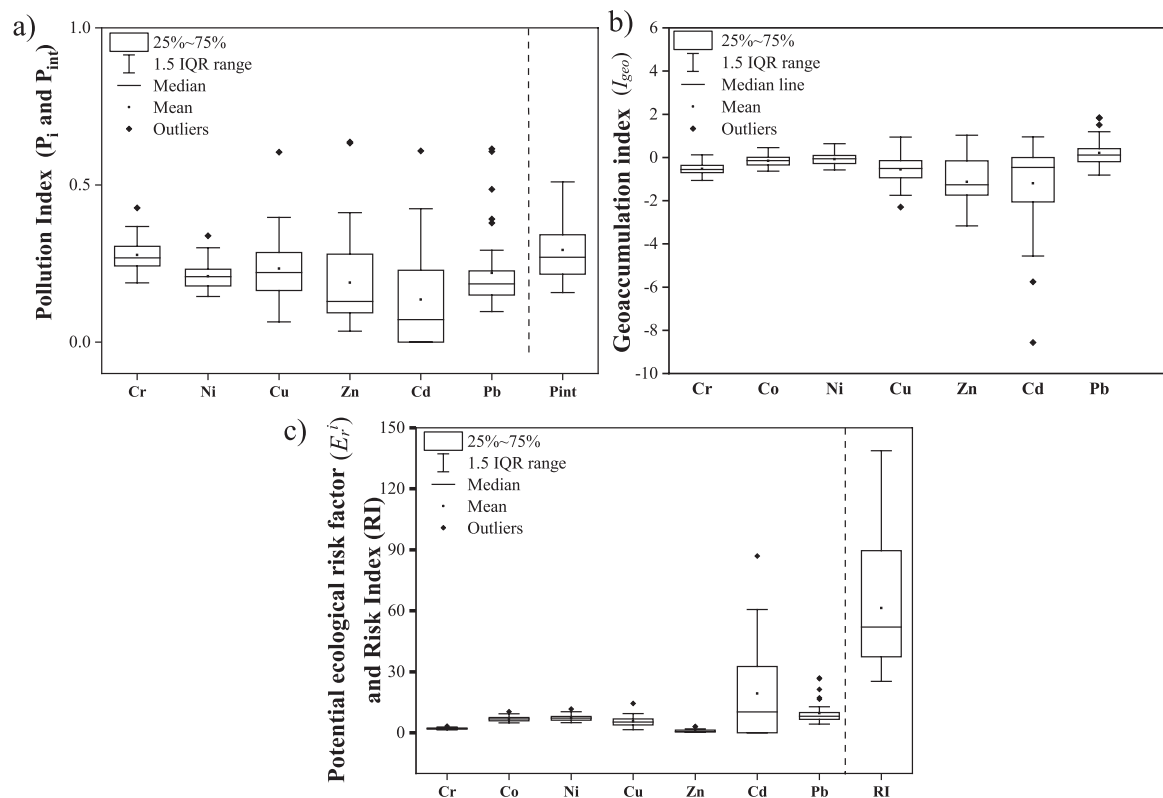


Fig. 4. Pollution and ecological risk assessment of heavy metals in surface soils from the oilfield-wetland zone of the Yellow River Delta: a) single-factor pollution index (P_i) and Nemerow composite pollution index (P_{int}); b) geoaccumulation index (I_{geo}); and c) potential ecological risk coefficient (E_r) and integrated ecological risk index (RI).

although differences were observed among wetlands [65]. Additionally, the proportions of I_{geo} values within the 0-1 range for Ni, Co, Cu, and Cd were 38.5%, 28.2%, 15.4%, and 12.8%, respectively, indicating localized mild enrichment of these metals at some sites.

The potential ecological risk coefficients and integrated ecological risk index are shown in Fig. 4c). The individual ecological risk coefficients followed the order $Cd > Pb > Ni > Co > Cu > Cr > Zn$. According to the classification criteria, the E_r^i values of most metals were below 40, indicating slight ecological risk, whereas Cd exceeded 40 at several sampling points, suggesting localized moderate ecological risk associated with Cd. Nevertheless, RI values remained below 150 at all sampling points, with a maximum value of 132.06, indicating low overall ecological risk in the study area. These results indicate that the overall ecological risk of heavy metals in the oilfield-wetland soils was low under the adopted risk assessment framework, although localized Cd-related risk should be monitored. Actually, the risk screening values for soil contamination of agricultural land applied in this study (GB 15618-2018) were applied as thresholds below which risks to agricultural product quality, crop growth, and soil ecological environments are considered low and generally negligible. The comparison between measured concentrations and these screening values, together with the ecological risk indexes, could provide a statutory benchmark for soil ecological risk under the current national regulatory framework. However, the soil conditions, such as shallow water tables and reducing redox states in the studied oilfield-wetland zones, may differ from those of typical cultivated agricultural lands. Therefore, the results should be interpreted cautiously regarding the ecological risks in this particular ecological setting.

Conclusions

This study investigated the concentrations, spatial distribution, potential sources, and ecological risks of Cr, Co, Ni, Cu, Zn, Cd, and Pb in surface soils from oilfield-wetland zones of the Yellow River Delta. The concentrations of all investigated metals with available Chinese agricultural soil risk-screening values were below their corresponding thresholds, indicating no regulatory risk under the current soil risk-control framework. However, Ni exceeded the local background value in all samples, while Pb, Co, Cr, and Cu frequently exceeded their corresponding background values, suggesting limited enrichment of several metals in the study area.

Spatial variation was limited for most metals. Cu differed significantly between Gudao Town and Xianhe Town, while Zn concentrations tended to be higher at sites farther from oil wells and in vegetated zones. Correlation analysis and PCA suggested that metal accumulation in the oilfield-wetland zone may reflect the

influence of natural pedogenic processes, with possible contributions from petroleum exploitation and other anthropogenic activities, such as traffic and agricultural disturbances. Pb showed a composite source pattern, whereas Zn appeared to be more closely associated with non-petroleum anthropogenic inputs.

The P_i , Nemerow index, I_{geo} , E_r^i , and RI results indicated generally low contamination and ecological risk in the study area. Nevertheless, Pb enrichment and consistent Ni background exceedance warrant continued monitoring, while elevated Cd-related risk was limited to a small number of sampling sites. Overall, this study provides baseline information for soil environmental management and ecological conservation in oilfield-wetland zones of the Yellow River Delta. However, it should be noted that this study only focused on surface soils (0-20 cm) to indicate the regional pollution status and ecological exposure to heavy metals, ignoring deeper soil layers where natural pedogenic background and historical accumulation trends might be recorded. Consequently, the vertical migration pathways and long-term sequestration dynamics of these metals remain unaddressed, which should be prioritized in future soil profile investigations. Source identification in this study was mainly based on correlation analysis and PCA, which are qualitative statistical approaches and cannot quantitatively determine the contribution rates of different sources, such as petroleum exploitation, traffic emissions, agricultural activities, and natural pedogenic processes. Future studies should combine PCA with quantitative source apportionment models, such as PMF or UNMIX, to better distinguish the relative contributions of different sources to heavy metal accumulation in oilfield-wetland zones.

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edited, and validated by the authors, who assume full and exclusive responsibility for the final content, scientific integrity, and originality of this work. The raw data supporting this study are available from the corresponding author upon reasonable request.

Conflict of Interest

The authors declare no conflict of interest.

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